THEUNIVERSITYOFLAHORE

Department of Computer Science & IT
Faculty of Information Technology

F-inal Year Project Proposal

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PRODUCT TITLE:	An AI-Powered Career Coaching and Job Preparation Platform
KEY WORDS:	AI, Career Coaching, Resume Builder, Cover Letter Generator, Mock Interviews, Job Preparation, Gemini API,WebRTC, Next.js, Vapi, Clerk, Prisma, PostgreSQL
DOMAIN OF THE PRODUCT:	Career Development / Artificial Intelligence / Web Application
SUPERVISOR'S NAME:	Taimur ul hassan

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1- Problem Identification:

Job preparation is a fragmented process, forcing candidates to use separate tools for resumes, cover letters, and interview practice, which often yields generic results.

1- Product Market Analysis:

The target audience for this project includes specific user classes:

- **Students and Fresh Graduates:** This group is tech-savvy but may lack knowledge of professional hiring standards.
- **Career Switchers:** These users are transitioning to a new field and need help reframing their existing skills.
- **Mid-Career Professionals:** Experienced job seekers looking to refine their application materials for senior roles.

2- Unique Value Proposition:

ZenAI is an integrated, AI-powered career coaching platform designed to streamline the entire application workflow. It turns scattered tasks into a single, efficient workflow. By unifying these critical tasks, ZenAI provides a seamless and efficient experience, empowering job seekers to build confidence and improve their employability.

3- Features:

The overall features to solve the problem are:

- **User Authentication and Profile Management:** Secure sign-up and login functionality managed by Clerk, where users can manage their personal and professional details.
- **AI-Powered Resume Enhancer:** Users can create or enhance a resume. The system analyzes it against a target job description and suggests quantified, impactful rewrites.
- **Tailored Cover Letter Generator:** Creates a focused and personalized cover letter that aligns the user's résumé with the job's requirements.
- **Dynamic Question Bank:** Generates a list of technical and behavioral interview questions based on the job role and required skills.
- **Realistic Voice-Based Mock Interviews:** Utilizes the Vapi service to conduct live, conversational mock interviews where an AI agent asks questions.
- **Interactive Screen Sharing & Code Analysis**: During technical mock interviews, users can share their screen. The AI (using vision capabilities) analyzes the user's code editor or presentation slides in real-time to provide specific feedback on coding syntax, logic, or presentation style.
- **Performance Feedback and Scoring:** After a mock interview, the system provides a detailed report with scores on communication, technical knowledge, and tips for improvement.
- **User Dashboard:** A central hub for users to view their generated documents, track interview history, and review progress.

4- Revenue Generation:

The project's financial viability is considered for future extension. The initial release will not include payment gateways for subscriptions , but the platform is being built as a robust and scalable application that can be extended with premium features in the future.

5- Marketing and Sales Strategy:

Mid-career professionals are noted as a "favored user class" as their success can lead to strong testimonials. Pricing models, such as subscriptions, are not part of the initial release but are considered for future development.

6- Risk Assessment:

Potential risks and challenges associated with this project include:

- **Third-Party API Dependency (Technical Risk):** The core functionality is heavily dependent on the availability and rate limits of external services like Gemini, Vapi, and Clerk. The design must account for potential API failures or latency.
- **Quality of AI Output (Usability Risk):** The platform depends on the quality and accuracy of the content generated by the Gemini API. While prompts are engineered for high quality, the output is not guaranteed to be perfect and may require user editing.
- **Data Privacy (Operational Risk):** The system must adhere to strict data privacy policies, storing only necessary user data and ensuring all sensitive information is securely managed.

7- Scalability and Growth Potential:

The scalability and growth potential of the project are analyzed as follows:

- **Scalable Technology:** The system is built using "modern, scalable, and serverless-friendly technologies". The backend API routes are designed to be stateless to allow for horizontal scaling and high availability. The database schema is designed to handle increasing amounts of data and user traffic.
- **Growth Opportunities:** Expansion opportunities for future development include direct integration with Applicant Tracking Systems (ATS), payment gateways for subscriptions, multi-tenant features for enterprise use, and connections to recruiter networks.

- **Business Goal:** A primary business goal is to build a "robust and scalable web application" that can serve a growing user base and be extended with premium features in the future.

8- Implementation and Execution Plan:

- **Technology Stack:** The system must be built using the specified stack: Next.js, Neon PostgreSQL, Prisma, Clerk, Vapi, WebRTC and Google Gemini API.
- **Architecture:** It is a full-stack application built with Next.js (for both frontend and backend) and a Neon PostgreSQL database managed via the Prisma ORM.
- **Core Services:** The system leverages the Google Gemini API for intelligent content generation , Clerk for secure user authentication , and the Vapi AI service for live, conversational mock interviews.
- **Deployment:** The deployment diagram shows a cloud infrastructure involving Vercel (to host the Next.js application) , Neon Cloud (for the PostgreSQL Database) , Google Cloud Platform (for the Gemini API) , Vapi Cloud Infrastructure (for the Vapi AI Service) , and Clerk Cloud Infrastructure (for the Authentication Service).

Supervisor's Signature: ………………………

Suggested Improvements regarding FYP (*if any*):